NATIONAL ECONOMICS UNIVERSITY, HANOI
COLLEGE OF TECHNOLOGY
FACULTY OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE



REPORT

Topic:

Optimizing Diabetes Risk Screening for Capacity-Constrained Healthcare Systems Using Boosting Models

---

Instructor: NGUYEN TRONG NGHIA

Student:	Hoang Viet Hung	11230749
	Le Phuong Khanh	11230750
	Ninh Thi Minh Thuy	11230779
	Chu Viet Anh	11230730

Hanoi, November 2026

Contributions

Table 1: Member Contributions

Member	Contribution Description
Chu Viet Anh	• Developed the continuous learning framework, including model monitoring, retraining, and governance. • Conducted temporal validation and evaluated model performance following retraining. • Developed the research questions and defined the study's research objectives.
Hoang Viet Hung	• Analyzed the dataset and exploratory findings. • Led model development, hyperparameter optimization, and experimental evaluation. • Compared global and capacity-specific XGBoost models using PR-AUC, Precision@K, Recall@K, and Lift@K. • Analyzed model performance and benchmarked results against Highest-k Loss.
Le Phuong Khanh	• Designed and implemented the deployment framework, including model interpretability and capacity-based patient prioritization. • Developed the Streamlit screening prototype and its integration with prediction logging. • Contributed to the study's limitations, practical implications, and conclusions.
Ninh Thi Minh Thuy	• Contributed to the research background, literature review, and methodological framework. • Developed the research questions and defined the study's research objectives. • Synthesized the research findings into the final report.

Abstract

Limited healthcare resources require diabetes screening systems to prioritize individuals for follow-up rather than rely solely on conventional classification performance. This study develops and evaluates a capacity-constrained screening framework using 253,680 observations and 21 predictors from the 2015 CDC Behavioral Risk Factor Surveillance System (BRFSS). After benchmarking Logistic Regression and screening four boosting algorithms, XGBoost was selected to investigate two optimization strategies: **Global Optimization**, which maximizes PR-AUC to produce a single model for all capacity levels, and **Capacity-Specific Optimization**, which maximizes Precision@K through six separately optimized models at capacity levels of 0.5%, 1%, 2.5%, 5%, 10%, and 20%.

Using stratified cross-validation and a locked test set, the study evaluates model performance through Precision@K, Recall@K, and Lift@K. The global XGBoost model achieved Precision@K values of **0.6614, 0.5416, and 0.4773** at the 1%, 5%, and 10% capacity levels, respectively. Capacity-specific optimization did not consistently outperform the global model, which also achieved competitive results against the published Highest-k Loss benchmark, although the experimental conditions differed.

These findings informed the decision to select the **global XGBoost model for deployment**, as it delivered comparable or superior performance across most capacity levels while avoiding the complexity of maintaining six specialized models. The selected model was integrated into a Streamlit research prototype supporting SHAP-based individual explanations, patient risk ranking, and capacity-based follow-up selection. Additionally, an evaluation using BRFSS 2025 data explored model monitoring and retraining, demonstrating modest improvements while highlighting remaining validation challenges.

Overall, the findings suggest that **a single globally optimized model can support flexible, resource-aware prioritization without the operational complexity of maintaining separate capacity-specific models**. The framework provides a foundation for screening decision support, although further validation is required before clinical application.

Keywords: Diabetes screening; XGBoost; Capacity-constrained healthcare; Precision@K; Patient prioritization; Machine learning.
Code Repository: GitHub Repository

Contents

Contributions 1

Abstract 2

1 Introduction 4
- 1.1 Background and Problem Context . . . 4
- 1.2 Research Gap . . . 4
- 1.3 Research Objectives . . . 4
- 1.4 Research Questions . . . 5
- 1.5 Research Contributions . . . 5

2 Literature Review 6
- 2.1 Diabetes Risk Prediction Using Machine Learning . . . 6
- 2.2 Boosting Models for Diabetes Prediction . . . 6
- 2.3 Capacity-Constrained Screening and High-Risk Prioritization . . . 6
- 2.4 Precision@K and High-Risk Patient Prioritization . . . 6
- 2.5 Highest-k Loss and Related Approaches . . . 7
- 2.6 Research Gap and Conceptual Positioning . . . 7

3 Material & Methodology 8
- 3.1 Data Understanding . . . 8
- 3.2 Research Design . . . 8
- 3.3 Methodology . . . 9
  - 3.3.1 Data Exploration, Cleaning, and Preparation . . . 9
  - 3.3.2 Validation Strategy, Baseline, and Model Screening . . . 9
  - 3.3.3 Global and Capacity-Specific XGBoost Optimization . . . 9
  - 3.3.4 Evaluation Framework and Model Comparison . . . 10
  - 3.3.5 Final Holdout Evaluation and Model Explainability . . . 10
- 3.4 Experimental Setup and Capacity Levels . . . 10

4 Results and Discussion 12
- 4.1 Exploratory Findings . . . 12
  - 4.1.1 Statistical Indicators . . . 12
  - 4.1.2 Distribution and Exploratory Associations . . . 12
- 4.2 Baseline and Model Screening Results . . . 13
- 4.3 Model Optimization and Final Performance . . . 13
  - 4.3.1 Global and Capacity-Specific Optimization . . . 13
  - 4.3.2 Final Locked-Test Evaluation . . . 13
  - 4.3.3 Comparison with Highest-k Loss . . . 14
- 4.4 Interpretation, Discussion, and Practical Implications . . . 14

5 Deployment Framework 16
- 5.1 Deployment Strategy . . . 16
- 5.2 Prioritization Workflow . . . 16
- 5.3 Integration . . . 16
- 5.4 Monitoring and Retraining . . . 17

6 Limitations and Conclusion 18
- 6.1 Limitations . . . 18
- 6.2 Conclusion . . . 18

1 Introduction

1.1 Background and Problem Context

Diabetes risk is associated with a wide range of health, behavioral, demographic, and socioeconomic factors, many of which are captured in population surveys such as BRFSS [1]. Survey-based indicators offer a low-cost basis for preliminary risk screening because they can capture lifestyle, health history, demographics, socioeconomic status, and healthcare access without requiring laboratory testing.

This study uses the CDC Diabetes Health Indicators dataset derived from the 2015 BRFSS, containing 253,680 observations and 21 predictors. The model is intended not to replace clinical diagnosis, but to support early prioritization of individuals who may require further assessment.

A key challenge is that healthcare providers cannot realistically follow up every individual identified as high risk because screening and intervention capacity is limited [2]. Under these constraints, the problem shifts from conventional classification toward **risk ranking and resource-aware patient prioritization**. For example, one provider may only be able to follow up the top 1% of predicted high-risk individuals, while another may accommodate 10% or 20%.

Accordingly, this study evaluates machine-learning models using capacity-based metrics such as Precision@K and investigates whether capacity-specific optimization provides meaningful advantages over a single globally optimized model.

1.2 Research Gap

Previous studies show that diabetes risk can be predicted from BRFSS health and behavioral indicators using machine-learning models, including boosting methods [2]. However, most evaluations rely on global metrics such as ROC-AUC, accuracy, precision, and recall, which do not directly reflect healthcare settings where only a limited proportion of patients can receive follow-up.

This study therefore focuses on **capacity-constrained screening**, where healthcare capacity is represented by the top-K proportion of patients selected for further assessment. A key unresolved question is whether optimizing a separate model for each capacity level provides meaningful benefits over using one globally optimized model across all capacities. This study directly compares these two strategies.

A further gap concerns deployment. Practical healthcare use requires not only risk prediction, but also patient ranking, capacity-based selection, interpretable predictions, and periodic model updating. Accordingly, the proposed framework combines capacity-aware evaluation, optimization comparison, and an interpretable deployment pipeline.

1.3 Research Objectives

The primary objective is to develop and evaluate a machine-learning framework for diabetes risk screening under healthcare capacity constraints. Specifically, the study aims to:

- identify key health, behavioral, demographic, and socioeconomic factors associated with diabetes risk;
- develop an optimized model for individual-level diabetes risk scoring;
- evaluate performance across multiple capacity levels using Precision@K;
- compare capacity-specific optimization with a globally optimized model and assess practical deployment feasibility.

1.4 Research Questions

This study is guided by the following research questions:

Primary Research Question: How effectively can survey-based health and behavioral indicators be used to predict diabetes risk under different healthcare capacity constraints?

- **RQ1.** Which health, behavioral, demographic, and socioeconomic factors are most strongly associated with diabetes risk?
- **RQ2.** How effectively can machine-learning models identify high-risk individuals across different healthcare capacity levels?
- **RQ3.** Does capacity-specific optimization improve Precision@K compared with a globally optimized model?
- **RQ4.** How can the resulting model be translated into an interpretable and deployable screening workflow for capacity-constrained healthcare settings?

1.5 Research Contributions

Existing studies using BRFSS data have mainly focused on improving overall diabetes prediction through conventional machine-learning and ensemble methods [2]. Although these approaches can achieve strong predictive performance, they typically rely on population-level metrics and pay less attention to the operational reality that healthcare providers can only follow up a limited proportion of patients. This study therefore shifts the focus from general classification performance toward **capacity-constrained patient prioritization**.
The main contributions of this study are:

- **Capacity-aware evaluation.** The study evaluates diabetes risk ranking across multiple healthcare capacity levels using Precision@K, Recall@K, and Lift@K, rather than relying only on a fixed classification threshold or global metrics.
- **Direct comparison of optimization strategies.** Capacity-specific XGBoost models are optimized separately for each capacity level and compared with a single globally optimized XGBoost model. The locked-test results show that capacity-specific optimization does not consistently improve out-of-sample performance, suggesting that additional model specialization may not always be justified.
- **Evidence of global-model robustness.** The global XGBoost model achieved Precision@K values of **0.6614, 0.5416, and 0.4773** at the 1%, 5%, and 10% capacity levels, respectively. These results are slightly higher than the published Highest-k Loss reference values of **0.657, 0.539, and 0.472**, respectively. This comparison is treated as an external benchmark rather than a direct head-to-head experiment.
- **Practical and interpretable deployment.** The proposed workflow combines individual risk scoring, patient ranking, capacity-based selection, and SHAP-based explanations to support practical screening decisions [3]. It also provides a basis for future model monitoring and periodic retraining as new patient data become available.

Overall, the study contributes not only another diabetes prediction model, but also evidence that **a single globally optimized model can provide stable top-risk identification across multiple capacity levels while avoiding the operational complexity of maintaining separate capacity-specific models**.

2 Literature Review

2.1 Diabetes Risk Prediction Using Machine Learning

Machine learning has been widely applied to diabetes risk prediction because it can combine demographic, behavioral, and health-related indicators. Using 2014 BRFSS data, Xie et al. [4] compared several algorithms and reported AUC values between **0.7182 and 0.7949**, showing that different models offer different trade-offs in discrimination and sensitivity.

More recent studies confirm that diabetes risk can be predicted using non-laboratory and survey-based information. Variables such as age, BMI, hypertension, cholesterol, general health, and physical activity are frequently identified as important predictors. BRFSS-based studies have also found strong performance from tree-based ensemble and boosting methods, with gradient boosting achieving approximately **0.80 AUROC and 0.45 PR-AUC** in one comparative analysis [5].

Overall, prior evidence supports machine learning for low-cost population screening, but no algorithm is universally superior. Performance depends on the dataset, preprocessing, class distribution, and evaluation objective, motivating empirical model comparison in this study.

2.2 Boosting Models for Diabetes Prediction

Boosting methods are well suited to diabetes prediction because they can capture nonlinear relationships and interactions among heterogeneous health indicators. XGBoost, introduced by Chen and Guestrin [6], is a scalable tree-boosting framework with regularization and efficient optimization, making it suitable for structured health-survey data.

Empirical studies have frequently reported competitive performance from boosting models. XGBoost achieved strong discrimination in survey-based diabetes prediction [5], while other studies using NHANES and BRFSS data also found XGBoost and related ensemble methods to perform well [7, 5]. However, more complex models do not always produce clinically meaningful gains, and some comparative studies emphasize the importance of interpretability and stability alongside predictive performance [5].

Accordingly, this study treats boosting as a strong candidate rather than an automatically superior approach and compares several boosting algorithms before deeper optimization. For interpretability, SHAP is used to explain both global feature importance and individual predictions [3].

2.3 Capacity-Constrained Screening and High-Risk Prioritization

Conventional disease-risk models are typically evaluated across the full population, whereas healthcare providers often have sufficient resources to follow up only a limited number of individuals. Under such constraints, the practical objective is not simply to classify every patient correctly, but to rank patients so that those with the highest predicted risk receive priority for further assessment. Yang et al. [2] describe this as a resource-constrained patient-prioritization problem, where screening capacity can be represented by the top proportion of patients selected for intervention.

This distinction is important because a model with strong overall discrimination may not necessarily identify the most relevant patients within a small actionable subgroup. Capacity-aware evaluation therefore focuses on performance at specific operating levels, such as the top 1%, 5%, or 10% of predicted risk. Such an approach more closely reflects real healthcare decisions, where available staff, diagnostic tests, and follow-up resources determine how many patients can realistically be screened.

2.4 Precision@K and High-Risk Patient Prioritization

Precision@K is a ranking-oriented metric that measures the proportion of true positive cases among the top K observations ranked by a model [8]. In healthcare screening, K can represent the number or percentage of patients that a provider is able to follow up. This makes Precision@K operationally different

from global measures such as ROC-AUC, which assess discrimination across the entire population rather than performance within the subgroup that will actually receive intervention.

A higher Precision@K indicates that limited screening capacity is concentrated on patients who are more likely to have the target condition. However, a trade-off arises as capacity increases: smaller values of K typically produce a more concentrated high-risk group, while larger values allow more positive cases to be captured but may include additional false positives. Yang et al. [2] therefore emphasize evaluation at relevant capacity levels. Precision@K can also be considered alongside Recall@K, which measures the proportion of all positive cases captured, and Lift@K, which compares the positive rate within the selected group with the underlying population prevalence.

2.5 Highest-k Loss and Related Approaches

Standard binary cross-entropy minimizes prediction error across all observations and does not directly emphasize performance among patients ranked at the highest risk. Focal Loss addresses a related issue by reducing the contribution of well-classified examples and assigning greater importance to difficult observations, particularly in imbalanced datasets [9]. However, difficult-to-classify observations are not necessarily the same as patients occupying the highest-risk positions in the ranking.

Yang et al. [2] proposed Highest-k Loss to align model training more directly with limited-capacity intervention. Rather than treating all observations equally, the method places greater optimization emphasis on cases receiving the highest predicted risk scores, with the objective of improving precision within the actionable top-K group. Because exact ranking and sorting operations are not differentiable, Highest-k Loss incorporates a differentiable sorting approach based on NeuralSort, allowing ranking information to be included in gradient-based model training [10].

On the 253,680-observation diabetes dataset, Highest-k Loss reported improvements in precision of approximately 0.05, 0.03, and 0.02 at the top 1%, 5%, and 10% risk levels, respectively, relative to conventional loss functions [2]. These approaches illustrate two alternative ways of incorporating capacity constraints: modifying the training loss itself, as in Highest-k Loss, or retaining a conventional predictive model while directly optimizing and evaluating performance using Precision@K at predefined capacity levels.

2.6 Research Gap and Conceptual Positioning

The literature establishes three key findings: survey-based indicators can support diabetes risk prediction without laboratory measurements; boosting models such as XGBoost are strong candidates for structured health data; and capacity-aware methods can improve prioritization when healthcare resources are limited.

However, it remains unclear whether healthcare systems require a separately optimized model for each operating capacity. Capacity-specific models may improve top-K performance, but they also increase tuning, validation, monitoring, and deployment complexity. A single global model may already provide sufficiently stable rankings across multiple capacities.

This study addresses that question by first selecting and optimizing a global XGBoost model using PR-AUC, then developing capacity-specific XGBoost models optimized directly for Precision@K. Keeping the model family constant allows the comparison to focus on the effect of the optimization objective.

The study therefore contributes by testing whether capacity-specific optimization provides generalizable gains over a global model and whether any improvement is large enough to justify additional operational complexity. Highest-k Loss is used as an external benchmark at comparable capacity levels.

3 Material & Methodology

3.1 Data Understanding

The study uses the CDC Diabetes Health Indicators dataset, derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS). The dataset contains 253,680 observations, 21 predictor variables, and one binary target indicating diabetes status. The predictors cover health conditions, health behaviors, demographic characteristics, socioeconomic status, and healthcare access, making the dataset suitable for population-level diabetes risk screening based largely on non-laboratory information.

Table 2: Dataset Overview

Aspect	Description
Data source	CDC Behavioral Risk Factor Surveillance System (BRFSS), 2015
Observations	253,680
Predictors	21
Target	`Diabetes_binary`
Target type	Binary classification
Continuous variables	BMI, MentHlth, PhysHlth
Ordinal variables	GenHlth, Age, Education, Income
Remaining predictors	Binary health, behavioral, demographic, and access indicators
Missing values	None detected
Positive class	35,346 diabetic observations
Negative class	218,334 non-diabetic observations
Diabetes prevalence	13.93%

The dataset is moderately imbalanced, with **86.07% non-diabetic** and **13.93% diabetic** observations, supporting the later use of ranking-oriented and positive-class-sensitive evaluation metrics. The predictors include binary health and behavioral indicators, ordinal variables such as general health, age, education, and income, and continuous variables including BMI, MentHlth, and PhysHlth. Exploratory analysis showed the strongest positive associations with diabetes for **GenHlth, HighBP, BMI, DiffWalk, and HighChol**, while income, education, and physical activity showed weaker negative associations.

The continuous variables are also non-normal. BMI is right-skewed, while MentHlth and PhysHlth are highly concentrated at zero. IQR-based potential outliers accounted for approximately **3.88% of BMI, 14.27% of MentHlth, and 16.14% of PhysHlth**, but all remained within documented valid ranges and were retained.

3.2 Research Design

This study adopts a **quantitative, predictive, and comparative research design** to evaluate diabetes risk screening under different healthcare capacity constraints. The analysis uses the CDC Diabetes Health Indicators dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), with diabetes status treated as a binary prediction target and 21 health, behavioral, demographic, and socioeconomic indicators used as predictors.

The research follows four main stages. First, exploratory data analysis assesses variable distributions, class imbalance, missing values, duplicates, outliers, and feature relationships. Second, the data are split into an 80% training set and 20% locked test set, with five-fold stratified cross-validation used during development. Logistic Regression provides the baseline, followed by screening and light tuning of HistGradientBoosting, XGBoost, LightGBM, and CatBoost.

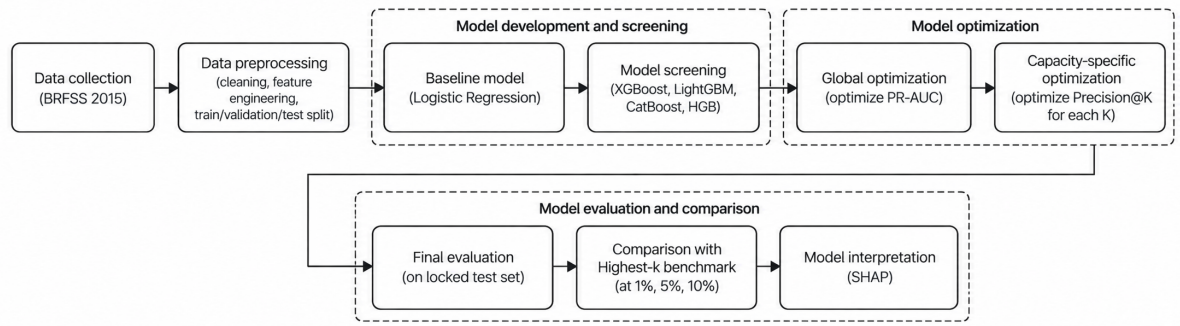


Figure 1: Research Process

Third, two XGBoost strategies are compared: a global model optimized for PR-AUC and six capacity-specific models optimized for Precision@K at 0.5%, 1%, 2.5%, 5%, 10%, and 20% capacity. Finally, all models are evaluated on the locked test set using PR-AUC, ROC-AUC, Precision@K, Recall@K, and Lift@K. Results at 1%, 5%, and 10% are additionally compared with Highest-k Loss benchmarks. SHAP and feature importance are used to support global and individual model interpretation.

3.3 Methodology

3.3.1 Data Exploration, Cleaning, and Preparation

The first stage examined dataset structure, class distribution, variable distributions, feature relationships, and data-quality issues. Spearman correlation was used because several variables are ordinal or non-normally distributed. No missing or invalid-range values were found. Identical rows were retained because no respondent identifier was available to confirm true duplicates. IQR-based outliers in BMI, MentHlth, and PhysHlth were also retained because they remained within valid survey ranges. For modeling, `Diabetes_binary` was used as the target and the remaining 21 variables as predictors. Standardization was applied only to Logistic Regression, while boosting models used the original feature scales.

3.3.2 Validation Strategy, Baseline, and Model Screening

The dataset was split into 80% training and 20% locked test data, stratified by diabetes status. All model development used five-fold Stratified K-Fold cross-validation, while the test set remained untouched until final evaluation.

An untuned Logistic Regression model served as the baseline. Four boosting models—HistGradientBoosting, XGBoost, LightGBM, and CatBoost—were then screened under the same validation framework using PR-AUC, ROC-AUC, and Precision@K.

LightGBM and XGBoost produced the strongest initial results and were retained for light comparative tuning, including adjustments to learning rate, tree complexity, boosting rounds, and class-imbalance handling. XGBoost achieved slightly better overall performance and was therefore selected for all subsequent optimizations. Using the same model family for both global and capacity-specific strategies ensured that later comparisons reflected differences in optimization objectives rather than differences between algorithms.

3.3.3 Global and Capacity-Specific XGBoost Optimization

XGBoost was optimized under two strategies. The global model used Optuna to maximize mean five-fold PR-AUC. A 30-trial pilot study was followed by a 100-trial full search after the initial results suggested that the boosting-iteration range should be expanded.

The capacity-specific strategy trained separate XGBoost configurations for six operating capacities: 0.5%, 1%, 2.5%, 5%, 10%, and 20%. Each study used 30 Optuna trials and directly maximized mean five-fold Precision@K at its corresponding capacity.

All models were trained on the complete training folds; capacity awareness was introduced only through the hyperparameter objective. The model family, data, and cross-validation procedure were kept constant so that the comparison isolated the effect of global versus capacity-specific optimization. This differs from Highest-k Loss, where prioritization is incorporated directly into the training loss [2].

3.3.4 Evaluation Framework and Model Comparison

The evaluation framework considered both global ranking performance and capacity-constrained prioritization. PR-AUC was used as the primary global metric, with ROC-AUC as a supplementary measure. For capacity-based evaluation, Precision@K measured the proportion of diabetic cases among the top-K selected patients, Recall@K measured the share of all diabetic cases captured, and Lift@K compared Precision@K with overall diabetes prevalence.

The main comparison was between a global XGBoost model optimized for PR-AUC and capacity-specific XGBoost models optimized for Precision@K, allowing the study to test whether separate models provide meaningful gains over one global ranking model.

3.3.5 Final Holdout Evaluation and Model Explainability

The locked test set was accessed only after all model and hyperparameter decisions were finalized. The global and capacity-specific XGBoost models were compared across all six capacity levels using Precision@K, Recall@K, and Lift@K.

At 1%, 5%, and 10% capacity, results were also compared with the published Highest-k Loss benchmarks from Yang et al. [2]. Because the studies used different test samples and training procedures, this comparison was treated as an external reference rather than a direct head-to-head evaluation.

The final global XGBoost model was interpreted using feature importance and SHAP [3], including both global and individual-level explanations. SHAP values were used to explain model behavior, not causal relationships. This supports the intended use of the framework as a prioritization and decision-support tool rather than a replacement for clinical diagnosis.

3.4 Experimental Setup and Capacity Levels

All experiments were conducted in Python using scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, and SHAP. A fixed random seed of 42 was used throughout.

Table 3: Core Experimental Configuration

Component	**Configuration**
Train-test split	80% / 20% locked test
Cross-validation	5-fold Stratified K-Fold
Global objective	Mean CV PR-AUC
Global Optuna search	30-trial pilot + 100-trial full search
Capacity-specific objective	Mean CV Precision@K
Capacity-specific search	30 trials per capacity

The global Optuna search initially explored `n_estimators` from 200–1000. Because the pilot indicated promising configurations near the upper boundary, the full search expanded this range to 200–1600. For capacity-specific optimization, the range was narrowed to 800–1600 based on the global-search

results. Other tuned parameters included tree depth, learning rate, subsampling, regularization, and `scale_pos_weight`. Six healthcare capacity levels were evaluated: 0.5%, 1%, 2.5%, 5%, 10%, and 20%. These represent progressively broader follow-up scenarios rather than universal clinical standards. The 1%, 5%, and 10% levels were included specifically to support comparison with Highest-k Loss [2].

With 50,736 observations in the locked test set, the six capacities corresponded to approximately 254, 508, 1,269, 2,537, 5,074, and 10,148 patients. The different tuning budgets reflected computational considerations and the use of global-search findings to guide capacity-specific optimization. While the global model underwent a broader search, the six capacity-specific models used 30 trials each with a refined search space to balance computational cost and optimization effort.

The same data split, cross-validation procedure, random seed, and XGBoost model family were maintained across global and capacity-specific experiments to support comparability. Nevertheless, differences in tuning budgets and search spaces may have influenced performance. Therefore, the results reflect the implemented optimization strategies rather than the isolated effect of their optimization objectives.

4 Results and Discussion

4.1 Exploratory Findings

4.1.1 Statistical Indicators

The dataset contains 253,680 observations and 22 variables, with no missing values. BMI has a mean of 28.38 and median of 27, indicating moderate right-skewness with values extending to 98. Both MentHlth and PhysHlth have medians of 0, showing that at least half of respondents reported no unhealthy days during the reference period, although both variables have long upper tails reaching 30 days. GenHlth, Age, Education, and Income remain within their predefined ordinal ranges, with relatively limited dispersion.

Overall, the descriptive statistics indicate a mixture of continuous and ordinal variables with notably non-normal distributions in several health indicators.

4.1.2 Distribution and Exploratory Associations



Figure 2: Exploratory Data Analysis of the BRFSS Diabetes Dataset

The EDA reveals several patterns relevant to subsequent modeling. Panel A shows a clear class imbalance, with **13.9% diabetic** and **86.1% non-diabetic** observations, supporting the use of minority-sensitive evaluation metrics rather than accuracy alone.

Panels B–D show that BMI is concentrated around the mid-to-high 20s, while MentHlth and PhysHlth are strongly right-skewed, with medians of zero. Panel E identifies the strongest positive Spearman associations with diabetes for GenHlth ($\rho = 0.29$), HighBP (0.26), BMI (0.23), DiffWalk (0.22), and HighChol (0.20), while Income, Education, and PhysActivity show negative associations.

Panel F supports this pattern, showing generally less favorable health profiles among diabetic respondents. Panel G combines smoking, physical activity, fruit and vegetable intake, and heavy alcohol use into a healthy-lifestyle score. Differences in diabetes prevalence across score levels suggest that behavioral factors may be more informative when considered jointly.

Overall, the EDA supports multivariable modeling because no single predictor clearly separates the two groups. However, Panel G should be interpreted cautiously because cross-sectional BRFSS data cannot distinguish pre-diagnosis behavior from lifestyle changes adopted after diagnosis.

4.2 Baseline and Model Screening Results

Logistic Regression was used as the baseline, achieving a **PR-AUC of 0.4073** and **ROC-AUC of 0.8228**. Under the capacity-constrained setting, its Precision@K declined from **0.6197 at 0.5% capacity** to **0.3884 at 20%**, providing a direct benchmark for patient prioritization.

Four boosting models were then screened under the same validation framework. All exceeded the Logistic Regression baseline in PR-AUC, with LightGBM and HistGradientBoosting producing the strongest default results.

Table 4: Default Model Screening Results

Metric	Logistic Regression	XGBoost	CatBoost	HistGradient Boosting	LightGBM
PR-AUC	0.4073	0.4225	0.4254	0.4343	0.4346
ROC-AUC	0.8228	0.8260	0.8281	0.8301	0.8303

LightGBM and XGBoost were retained for light comparative tuning. After tuning, XGBoost achieved a slightly higher PR-AUC (**0.4372 vs. 0.4362**) and higher Precision@K at five of the six capacity levels.

Table 5: Precision@K After Light Tuning

Model	0.5%	1%	2.5%	5%	10%	20%
LightGBM	0.7054	0.6768	0.6271	0.5609	0.4902	0.4000
XGBoost	**0.7084**	**0.6857**	0.6248	**0.5617**	**0.4916**	**0.4007**

XGBoost was therefore selected for deeper global and capacity-specific optimization.

4.3 Model Optimization and Final Performance

4.3.1 Global and Capacity-Specific Optimization

The global XGBoost model was optimized using Optuna with mean five-fold PR-AUC as the objective. A 30-trial pilot search was followed by a 100-trial full search after the boosting-iteration range was expanded.

The final global configuration achieved a **mean cross-validation PR-AUC of 0.4389**, pooled out-of-fold PR-AUC of **0.4384**, and ROC-AUC of **0.8316**.

Capacity-specific models were then optimized separately at 0.5%, 1%, 2.5%, 5%, 10%, and 20% capacity using mean five-fold Precision@K. Best cross-validation Precision@K ranged from **0.7300 at 0.5%** to **0.4012 at 20%**. Although these models produced small validation improvements at several thresholds, the gains were limited.

4.3.2 Final Locked-Test Evaluation

All finalized models were evaluated once on the locked test set of **50,736 observations**.

Table 6: Final Locked-Test Performance

Capacity	Global P@K	Capacity P@K	Global Recall@K	Capacity Recall@K	Global Lift@K	Capacity Lift@K
0.5%	**0.6890**	0.6732	**0.0248**	0.0242	**4.94**×	4.83×
1%	0.6614	0.6614	0.0475	0.0475	4.75×	4.75×
2.5%	**0.6052**	0.6028	**0.1086**	0.1082	**4.34**×	4.33×
5%	**0.5416**	0.5388	**0.1944**	0.1934	**3.89**×	3.87×
10%	0.4773	**0.4781**	0.3426	**0.3432**	3.43×	3.43×
20%	**0.3966**	0.3963	**0.5694**	0.5690	**2.85**×	2.84×

Capacity-specific optimization did not consistently outperform the global model. The global model performed better at four capacities, tied at 1%, and was only marginally lower at 10%.

Precision decreased as capacity expanded, while Recall@K increased. Lift remained above **2.8**× across all capacities and reached nearly **5**× **at 0.5%**, showing that the models concentrated diabetic cases far more effectively than random selection.

4.3.3 Comparison with Highest-k Loss

Table 7: Comparison with Published Highest-k Results

Capacity	Highest-k P@K	Global XGBoost	Capacity-Specific
1%	0.657	**0.6614**	**0.6614**
5%	0.539	**0.5416**	0.5388
10%	0.472	0.4773	**0.4781**

The proposed models were competitive with the published Highest-k benchmark. The global model slightly exceeded the reported values at all three comparable capacities.

However, this should be interpreted as an external benchmark comparison, not a direct head-to-head evaluation, because the studies used different experimental splits and training procedures.

4.4 Interpretation, Discussion, and Practical Implications

The main result is that **capacity-specific optimization did not consistently outperform the global XGBoost model on the locked test set**. Although separate models achieved slightly higher cross-validation Precision@K at some thresholds, the global model performed better at four of six capacities, tied at 1%, and was only marginally lower at 10%. This suggests that a single PR-AUC-optimized model can provide a sufficiently stable ranking across different capacity levels.

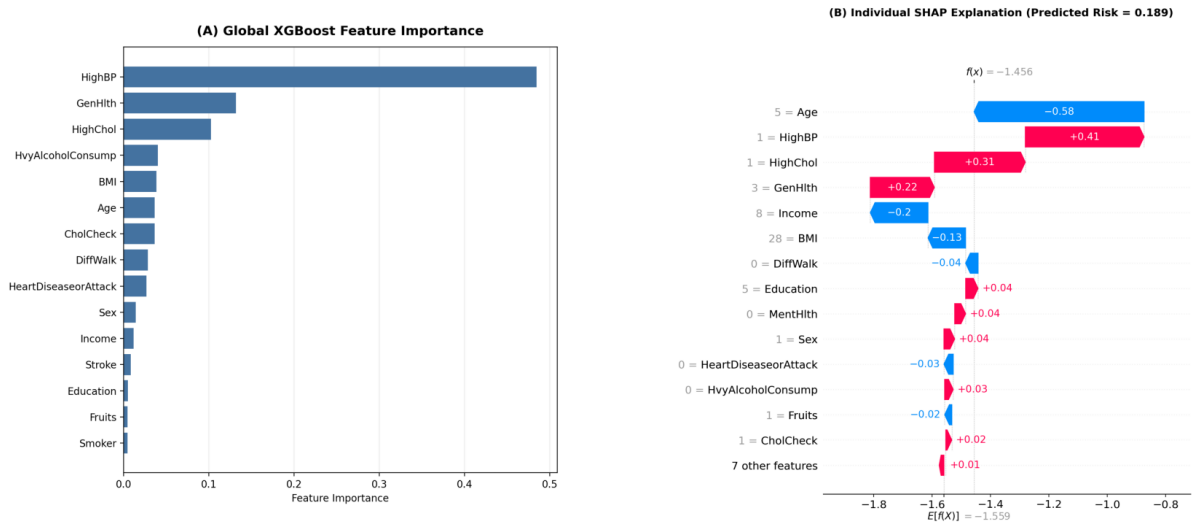


Figure 3: Global XGBoost Feature Importance and Individual SHAP Explanation

Figure 3 illustrates model interpretability at both global and individual levels. **Panel A** shows that high blood pressure (HighBP) is the dominant feature in the global XGBoost importance ranking, followed by general health (GenHlth) and high cholesterol (HighChol). Other health and demographic indicators, including BMI, age, and mobility difficulties, also contribute to model predictions.
Panel B presents a SHAP waterfall explanation for a selected test observation with a predicted diabetes probability of 0.189. High blood pressure, high cholesterol, and general health increase the predicted risk score, whereas age, income, and BMI contribute negatively for this individual. The SHAP contributions are expressed on the model's log-odds scale rather than directly as probability changes.
Together, these findings demonstrate how the model combines multiple health indicators while providing both overall feature rankings and individual-level explanations. Importantly, feature importance and SHAP values describe predictive associations rather than causal effects.

From a practical perspective, the findings support a simple deployment approach: use one global model to generate risk scores, rank patients, and select the top K according to available healthcare capacity. This avoids maintaining separate models for each capacity level while allowing the screening threshold to adapt to changing resources.

5 Deployment Framework

The selected global XGBoost model is presented as a research prototype linking survey-based status scoring to capacity-constrained follow-up prioritization (Figure 4).

5.1 Deployment Strategy

The prototype deploys one globally optimized XGBoost model through Streamlit Community Cloud, with source code maintained on GitHub. This choice follows the locked-test comparison: capacity-specific optimization did not consistently improve Precision@K. The application uses the same 21 BRFSS predictors as the trained model and does not retrain during inference.

The model distinguishes the survey-reported diabetes-status class recorded at the time of the BRFSS survey; it does not predict future diabetes incidence. Its output is therefore treated as a **relative screening score**, not a clinically validated probability or diagnosis. In single-patient mode, the app reports the score and top three SHAP contributors and logs the prediction to Google Sheets. Batch mode validates anonymous Excel/CSV records and exports ranked results; it does not log batch records. Outcome verification, monitoring, and retraining are proposed extensions that would require independent evaluation before any model replacement.

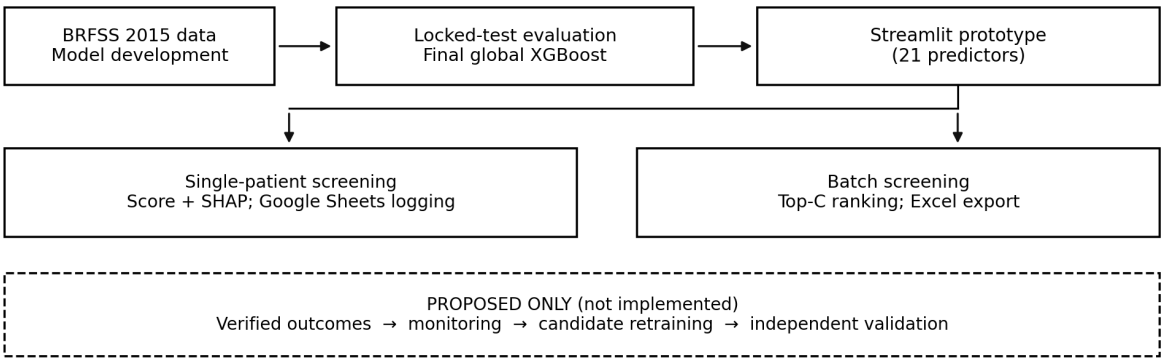


Figure 4: Model Development, Prototype Deployment, and Proposed Continuous-Learning Extension

5.2 Prioritization Workflow

For a batch of N valid records, a provider enters its available follow-up capacity C (patients). The global model scores and ranks all records; the first $m = \min(C, N)$ are selected, giving an effective capacity of $K = 100m/N$ (%). Ties are resolved using the original file order. Thus, 75 follow-up places for 1,000 records yield a **Top-7.5% shortlist** without changing models.

The exported workbook contains the complete ranking, selected subset, and methodology notes. The six historical Precision@K and Recall@K results describe the separate locked test set, not new uploads. Selection is intended to support preliminary follow-up planning, not automated clinical decisions.

5.3 Integration

The prototype separates model inference from updating. Streamlit Community Cloud serves a global XGBoost model from a versioned deployment bundle containing the portable model, 21-feature schema, SHAP reference data, risk-score quantiles, calibration evidence and capacity metrics. The application checks model-file integrity before loading and does not train during prediction. Single-patient inputs and outputs are sent to Google Sheets through a Google Apps Script endpoint, with a unique prediction ID

and model version; outcome fields remain blank until verified. Batch uploads are validated, ranked and exported as Excel, but are not logged to Google Sheets.

5.4 Monitoring and Retraining

An offline retraining pipeline was implemented and tested with BRFSS 2025 records, rather than outcomes collected from app users. BRFSS 2015 remains the original development cohort and locked-test reference; 2025 enables a later-period evaluation using the same 21 predictors. The 2024 public data lack several required blood-pressure, cholesterol and dietary indicators, making 2025 more suitable for the reported retraining experiment [11].

According to the pipeline report, **201,093 of 356,158 respondents in 2025 (56.5%)** had complete, recordable records. Of these, **160,934** joined the original 2015 training split (202,944 observations) to train candidate v1.1.0; the remaining **40,159** were reserved as an unseen temporal holdout. The 2015 locked test ($n = 50{,}736$) was retained to detect regression. The 2025 target was coded to match the 2015 file: reported diabetes = 1; prediabetes, diabetes only during pregnancy, and no diabetes = 0. Although the dataset documentation describes the 2015 target as "prediabetes or diabetes", its 35,346 positive cases (13.93%) correspond to reported diabetes only, supporting the intended alignment of the two targets. However, two considerations limit the interpretation of this temporal evaluation. First, only 56.5% of BRFSS 2025 respondents were retained after applying the completeness and feature-compatibility requirements. This may introduce selection bias if excluded respondents systematically differ from the retained population. Second, although the 2025 target was recoded to align with the diabetes-status definition used in the 2015 dataset, differences in survey questions and coding procedures may still affect comparability across years. Therefore, the temporal results should be interpreted as evidence of technical feasibility rather than definitive evidence of generalizability to the broader 2025 population.

The monitoring procedure validates feature values and outcome labels, checks for duplicate IDs and compares the deployed and candidate models using PR-AUC, ROC-AUC, Brier score and Precision@K across the six study capacities. A candidate is considered only if it improves PR-AUC on new data while staying within predefined 2015-test tolerances (PR-AUC decrease ≤ 0.005; Brier increase ≤ 0.002; Precision@K decrease ≤ 0.02 at each capacity). Model replacement still requires team approval.

Table 8: Summary of the Reported BRFSS 2025 Temporal-Holdout Results ($n = 40{,}159$)

Metric	Deployed v1.0.0	Candidate v1.1.0	Change
PR-AUC	0.4155	**0.4197**	+0.0042
ROC-AUC	0.8085	**0.8113**	+0.0028
Brier score	0.1085	**0.1080**	−0.0005
Precision@0.5%	0.6517	**0.6766**	+0.0249
Precision@10%	0.4771	**0.4858**	+0.0087

The candidate improved PR-AUC on the 2025 holdout (paired-bootstrap 95% interval for the difference: +0.0026 to +0.0063) and identified more positive cases within all six capacity groups. On the 2015 locked test, PR-AUC declined only from 0.4258 to 0.4247 and Brier score increased from 0.0987 to 0.0992, both within the stated tolerances.

Nevertheless, the gain on newer data was modest and did not fully recover the observed temporal decline. The run also bypassed the proposed 30-day waiting period and skipped cross-validation; its findings therefore support a **feasibility demonstration, not an automatic production update**.

6 Limitations and Conclusion

6.1 Limitations

Several limitations should be considered when interpreting the findings. First, the study relies on the 2015 BRFSS dataset, which contains self-reported, cross-sectional information. Consequently, the model identifies patterns associated with diabetes status at the time of the survey rather than predicting future disease onset. Survey responses may also introduce recall and reporting biases.

Second, although the locked-test evaluation supports the global model's ranking performance, its generalizability to newer populations and clinical settings remains uncertain. The model's probability calibration is imperfect, and performance across demographic and socioeconomic subgroups has not been systematically evaluated.

Third, the comparison with Highest-k Loss uses published results rather than identical experimental conditions, limiting direct conclusions about methodological superiority.

Finally, the Streamlit application demonstrates technical feasibility but has not undergone clinical validation or integration with real healthcare workflows. Further temporal validation, subgroup analysis, and outcome-based monitoring are necessary before practical adoption.

6.2 Conclusion

This study investigated diabetes screening under healthcare capacity constraints using 253,680 BRFSS observations and 21 predictors. Rather than focusing exclusively on conventional classification metrics, it evaluated how effectively machine-learning models prioritize individuals when follow-up resources are limited.

The globally optimized XGBoost model achieved Precision@K values of **0.6614, 0.5416, and 0.4773** at the 1%, 5%, and 10% capacity levels, respectively. Capacity-specific optimization did not consistently improve locked-test performance, suggesting that maintaining separate models for different capacities may offer limited practical benefits. The global model also produced competitive results against the published Highest-k Loss benchmark, although this comparison was not conducted under identical experimental conditions.

These findings support a simpler deployment strategy in which one global model generates screening scores and prioritizes individuals according to available follow-up capacity. The accompanying Streamlit prototype demonstrates this approach through interpretable predictions and capacity-based ranking.

Overall, the study highlights the value of aligning machine-learning evaluation with operational constraints. While further validation is required, the proposed framework provides a practical foundation for resource-aware, survey-based screening and prioritization.

References

[1] UCI Machine Learning Repository. *CDC Diabetes Health Indicators*. Data set. 2017. DOI: 10.24432/C53919. URL: https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators.

[2] Hongyi Yang et al. "A Practical Approach to Disease Risk Prediction: Focus on High-Risk Patients via Highest-k Loss". In: *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2023, pp. 3226–3233. DOI: 10.1109/BIBM58861.2023.10385816.

[3] Scott M. Lundberg and Su-In Lee. "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017, pp. 4765–4774. URL: https://papers.neurips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.

[4] Zidian Xie et al. "Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques". In: *Preventing Chronic Disease* 16, E130 (2019). DOI: 10.5888/pcd16.190109.

[5] Mustapha Aliyu Muhammad, Jamilu Sani, and Mohamed Mustaf Ahmed. "Exploring Explainable Machine Learning for Predicting and Interpreting Self-Reported Diabetes among Tennessee Adults: Insights from the 2023 Behavioral Risk Factor Surveillance System (BRFSS)". In: *Journal of Primary Care & Community Health* 16, 21501319251400546 (2025). DOI: 10.1177/21501319251400546.

[6] Tianqi Chen and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery, 2016, pp. 785–794. DOI: 10.1145/2939672.2939785.

[7] Keshab Raj Dahal et al. "Comparative Performance of Machine Learning Models for Diabetes Prediction Among US Adults Using NHANES Data". In: *International Journal of Statistics and Probability* 15.1 (2026). DOI: 10.5539/ijsp.v15n1p12.

[8] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008. DOI: 10.1017/CBO9780511809071.

[9] Tsung-Yi Lin et al. "Focal Loss for Dense Object Detection". In: *Proceedings of the IEEE International Conference on Computer Vision*. 2017, pp. 2980–2988. URL: https://openaccess.thecvf.com/content_iccv_2017/html/Lin_Focal_Loss_for_ICCV_2017_paper.html.

[10] Aditya Grover et al. "Stochastic Optimization of Sorting Networks via Continuous Relaxations". In: *International Conference on Learning Representations*. 2019. URL: https://openreview.net/forum?id=H1eSS3CcKX.

[11] Centers for Disease Control and Prevention. *2025 BRFSS Survey Data and Documentation*. 2026. URL: https://www.cdc.gov/brfss/annual_data/annual_2025.html.